



Rigid-Flex Deep Dive

Beyond the basic rigid-flex deck — production design + reliability

When you need to combine rigid PCBs with flex interconnects in one assembly

Audience: Engineers building wearables, cameras, drones

Stackup • Bend radius • Reliability • Cost • Manufacturing



Why Rigid-Flex (Recap)

Connector eliminate

Direct flex tail vs
cable + connectors → fewer
failure points.

3D fitment

Wrap around battery,
lens, mechanism.

Weight

No cable + connector hardware
= lighter assembly.

Vibration

Solid interconnect
survives shock + vibration.

Cost

3-5× rigid PCB cost
but net cheaper than
discrete cable + connectors.

Reliability

100k+ flex cycles
in dynamic apps.

Construction Detail

- ▶ Rigid sections: standard FR-4 with copper layers + dielectric
- ▶ Flex sections: polyimide film (Kapton) + rolled-annealed copper
- ▶ Layers continuous through transition (flex Cu extends INTO the rigid section)
- ▶ Bonded with adhesive + heat + pressure
- ▶ Coverlay on flex section (vs solder mask on rigid)
- ▶ Stiffeners under connector mounting points (separate piece bonded)
- ▶ Build-up: outer layers + inner core + flex layers laminated as one stack
- ▶ Layer count: 2-12+ layers possible (most products 4-6 layer rigid + 2-layer flex)

Rigid-Flex Stackup Examples

4-layer Rigid + 2-layer Flex

- ▶ L1 (top rigid): Signal
- ▶ L2 (top inner): GND plane
- ▶ L3 (bottom inner): PWR plane
- ▶ L4 (bottom rigid): Signal
- ▶ Flex layers: 2 copper (L1+L4 or L2+L3)
- ▶ Used in: wearables, small cameras

6-layer Rigid + 4-layer Flex

- ▶ L1: Signal
- ▶ L2: GND (continuous through flex)
- ▶ L3: Signal (flex carries 2 signals)
- ▶ L4: Signal (flex carries 2 signals)
- ▶ L5: PWR
- ▶ L6: Signal
- ▶ Used in: complex flex tails, drones

Bend Radius (Detail)



Application	Bend cycles	Bend radius (min)
Static (one bend, assembly only)	1 cycle	6× thickness
Static, multilayer	1 cycle	10× thickness
Static, multilayer 6+ layer	1 cycle	15× thickness
Dynamic (camera tilt, hinge)	100k cycles	100× thickness
Continuous dynamic	1M+ cycles	200× thickness
Robotic arm	10M+ cycles	200-500× thickness

For 0.2 mm flex thickness

Static 4-layer: bend radius ≥ 3 mm. Dynamic (camera): ≥ 20 mm. Continuous dynamic: ≥ 40 mm.

Flex Layout Rules (Detail)

- ▶ Traces PERPENDICULAR to bend axis — they take the least stress
- ▶ Avoid vias in the bend region (concentrate stress)
- ▶ Stagger traces on different layers — no vertical alignment in bend area
- ▶ Tear-drop pad junctions where traces meet pads
- ▶ Hatched ground plane (not solid) on flex — solid is too stiff to bend
- ▶ Service loops (extra length) — allow flex to move freely
- ▶ Curved (not 90°) traces in bend zones
- ▶ Stiffeners ONLY under connector mounting points + rigid sections

Stiffeners

- ▶ Stiffener = small piece of rigid material bonded under specific areas
- ▶ Common: FR-4 stiffener (matches rigid section material)
- ▶ Alternative: polyimide stiffener (thicker than coverlay)
- ▶ Or: stainless steel stiffener (for connector mating force)
- ▶ Placement: under SMD components on flex, under connector solder pads
- ▶ Stiffener thickness: 0.2-0.5 mm typical (1.6 mm for major connectors)
- ▶ Stiffener size: at least 2 mm larger than component footprint
- ▶ Cost: ~10-20% additional per stiffener

Reliability Testing

- ▶ Bend cycle test: motorised flex tester applies repeated bends
- ▶ Pass criterion: < 10% resistance change after N cycles
- ▶ Thermal cycle: -40 to +85 °C × 1000 cycles
- ▶ Humidity test: 85 °C / 85% RH for 1000 hours
- ▶ Vibration: random spectrum per MIL-STD-810
- ▶ Drop test: for portable products
- ▶ X-ray inspection at flex-to-rigid transition (most likely failure point)
- ▶ Microsection of flex-rigid joint at end-of-test for failure analysis

Manufacturing Process

- ▶ Specialty fabs only — not all PCB fabs do rigid-flex
- ▶ Process more complex than rigid: handle flex film + rigid layers separately, then laminate
- ▶ Coverlay application + cure on flex sections
- ▶ Drill + plate vias (especially careful in flex regions)
- ▶ Final laminate: rigid + flex bonded into one rigid-flex panel
- ▶ Routing: separate rigid and flex regions during panel cutting
- ▶ Yield: lower than rigid-only PCBs — quality control more demanding
- ▶ Indian rigid-flex fabs: AT&S India, Centum, specialty PCB suppliers

Cost + Lead Time



Configuration	Proto cost (5-10 pcs)	Volume cost (10k)	Lead time
2-layer flex only	₹3-8k	₹150-300/board	10-20 days
4-layer rigid + 2 flex	₹20-40k	₹800-1500/board	20-30 days
6-layer rigid + 4 flex	₹40-80k	₹1.5-3k/board	30-40 days
8-layer rigid + 6 flex (HDI)	₹80-150k	₹3-6k/board	40-50 days
Multilayer rigid-flex with stiffener	+10-20%	+10-20%	+5-10 days

Design Tools

- ▶ Altium Designer: best commercial rigid-flex support (sub-stack support)
- ▶ Cadence Allegro: enterprise tier, robust rigid-flex
- ▶ KiCad 8+: basic rigid-flex support — improving with each release
- ▶ Mentor Xpedition: enterprise-tier with rigid-flex 3D viewer
- ▶ Use proper rigid-flex CAD (not just adding a flex board to rigid file)
- ▶ Tool features: per-region stackup, bend zone visualization, 3D bent view
- ▶ Cost: Altium ₹3L+/year, KiCad free

Common Rigid-Flex Mistakes

- ▶ Bend radius too small → copper cracks during first assembly bend
- ▶ Vias in bend area → stress concentration → trace breaks under fatigue
- ▶ Solid Cu plane on flex → too stiff to bend, kinks instead
- ▶ Sharp 90° trace corners on flex → stress concentration
- ▶ Forgotten stiffeners under connectors → solder joints crack with insertion force
- ▶ Mixed bend radius requirements (some dynamic + some static) on same flex
- ▶ Wrong copper type — ED Cu used for dynamic flex → cracks within 100 cycles
- ▶ Specifying rigid-flex from a regular rigid PCB fab → can't deliver

Key Takeaways

Key takeaways from this lecture

Bend radius is sacred

Dynamic $100 \times t$ minimum.

Traces $\perp$ bend axis

Reduce copper stress.

Hatched ground on flex

Solid is too stiff.

Stiffeners under connectors

Anchor solder joints.

RA copper for dynamic flex

ED cracks early.

Specialty fab needed

Not your regular rigid shop.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ IPC-2223 — flex board design standard
- ▶ IPC-6013 — flex board performance specification
- ▶ DuPont Pyralux design guide
- ▶ Multech / Streamline Circuits — specialty fab resources
- ▶ Phil's Lab YouTube — rigid-flex video
- ▶ Altium rigid-flex tutorial
- ▶ AT&S India + Centum — Indian fab options

Thank you • Build something this week.